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Conceptual design and realization of a graphical tool for determining the test coverage of digital circuit diagrams

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vorgelegt von

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Abstract

According to Printed circuit board (PCB)s and their digital circuits which are getting more complex all the time there is the requirement of being able to determine the test coverage of it. In dependency on these determined values then automatic boundary scan chains could be performed or the need of changing something on the layout will be visible. This document describes the way of the conceptual design and the realization of a graphical tool for loading, showing different digital circuit diagrams projects which need to be in the Open database (ODB)++ format and the determination of test coverage parameters as well as the creation of Serial vector format (SVF) files for being able to run the boundary scan chain on these special devices which contains at least one Joint test action group (JTAG) device. You will get detailed information about the frameworks, programming patterns and best practices which were used for creating this Model view viewmodel (MVVM) C# project and the project structure itself which was created for making sure to have the following features available:

- Opening any ODB++ project by connecting to a stand alone Google remote procedure calls (GRPC) server and using the PCB-Investigator API
- Exporting any opened projects to Java script object notation (JSON) file for later usage and importing of that created files
- Creation and loading of project files which contains a manifest file for information about the project structure itself
- Modifying, exporting and importing of different kind of settings according the connection to the GRPC server, the appearance and the test coverage values
- Determination of different test coverage values according to the boundary scan mechanism
- Creation of SVF files for run the boundary scan chain with a programmer from ProMik on a JTAG containing device

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Chapter 1.

List of abbreviations

MES Manufacturing execution system

SAP Semi-automated production programming station

PCB Printed circuit board

ODB Open database

JTAG Joint test action group

SVF Serial vector format

GRPC Google remote procedure calls

JSON Java script object notation

MVVM Model view viewmodel

MCU Micro controller

GUI Graphical user interface

Chapter 2.

Introduction

Within this chapter i will describe the motivation, the requirements, the used programming language and operating system for creation of the whole project. All these aspects were previously discussed with the company to make sure to create a most valuable software for them.

2.1. Motivation

As our company is already providing programmers for being able to write data into different Micro controller (**MCU**)s now there came up the requirement, in dependency on customer demands, to perform boundary scan chains on **JTAG** devices. For being able to perform these kind of tests there should be a software available which allows to determine first the test coverage parameters in dependency on the boundary scan mechanism and create such **SVF** files for playing them with the programmer. This software should be provided as a complex framework where the opportunity is given to extend it in the future in elementary way by using much interfaces.

2.2. Requirements

In order to provide all features as needed there were a couple of requirements which popped up at the very beginning of the conceptual design and during the implementation of the software:

- The software itself should consist of a **GUI!** (**GUI!**) application and a separate **GRPC** server for using the **PCB**-Investigator API
- There should be the possibility of opening **ODB++** project folders and display the result of circuit diagram in a pretty way
 - There should be the possibility of moving the visible area by mouse movement and zooming in and out of the visible area

- The visibility of each component should be settable
 - All connections of one component to the others should be displayable
 - The whole view of all components could be mirrorable along both axes
- An export mechanism of an opened project into **JSON** file for later usage should be available as an according import mechanism to be independent from the **PCB-Investigator** API as this requires a separate license for usage
- Settings should be available, exportable and importable
 - General settings according the appearance of the loaded project view
 - **GRPC**-Server related connection settings
 - General settings according the request of **PCB-Investigator** API
 - Test coverage related settings for determining all components for the boundary scan chain
 - **SVF** related settings according the programmer connection parameters to use
- Determination of test coverage using the boundary scan mechanism
- Creation of **SVF** files containing all instructions to be performed by the programmer while playing the boundary scan chain on a **JTAG** compatible device

2.3. Programming language and operating system

Chapter 3.

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3.2. ODB++ project format

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3.4. Boundary scan

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Conclusion

Chapter 8.

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Appendix A.

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